

Lab 4 - Soft Inflatable Actuators

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I. OBJECTIVES AND EXPERIMENTAL SETUP

The purpose of this lab was to deepen our understanding of the fabrication process for inflatable actuators using a thermoplastic elastomer, TPE. We compared the functionality of these actuators to that of previously constructed silicone actuators, gaining insights to guide the construction of future designs.

We made three designs using the TPE elastomer to display its functionality, all using the same method of fabrication. The TPE plastic is cut and then heat sealed using a press. A hole is then punctured in the TPE and then tubing is inserted through it for inflation/deflation. The designs that we created can be seen in figures 1-3: a vine robot to display the potential stroke, a muscular robot to display the ability to apply forces, and a flower actuator to display other potential functionalities. The lifting force and stroke of the muscle actuator was characterized by evaluating the change in length of the robot as weights were attached to one end (figure 1). The work done per volume air was computed for the muscle as well by multiplying the applied weight by gravity and the change in length.

II. RESULTS & ANALYSIS

When the actuators are inflated, it is observed that the TPE does not stretch. Instead, for the flower actuator and the vine actuator, the material unfolds from inside itself. For the muscle actuator, the material expands. This is a form of geometric deformation, different from the previous lab where the silicone elastomer materials did stretch. Therefore, there is no change in pressure inside of the TPE actuator, unlike the elastomer actuators. The force of the TPE actuator is also within the same order of magnitude of the elastomeric actuator, as seen in Figure 4. When stretching TPE, the material elongates, and once released it either mostly returns to its original shape or it snaps. Over time with many stretches, the material begins to get worn out and deforms. This is an important observation because after repeated testing, TPE will begin to not reproduce test results.

It was noticed that the stroke of the vine actuator is able to double its collapsed length. This is a significant increase compared to the elastomeric actuators observed in lab 3, where their length increased by ~1%. However, the elastomeric actuators have a greater stroke than that of the TPE muscular actuator when supporting similar weights. We do not know the volume change of the flower since it is an irregular shape.



Figure 1: Muscle Actuator **Figure 2: Flower Actuator**
Figure 3: Vine Actuator

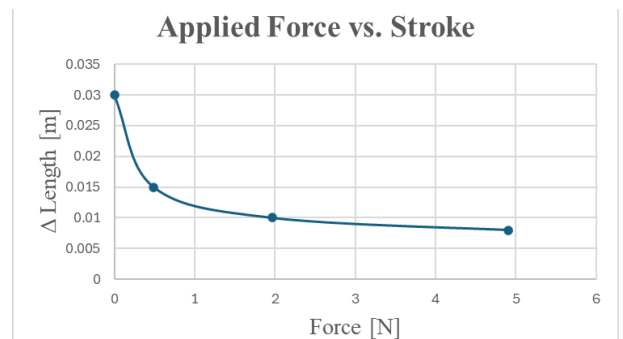


Figure 4: Applied Force & Stroke of Muscle Actuator

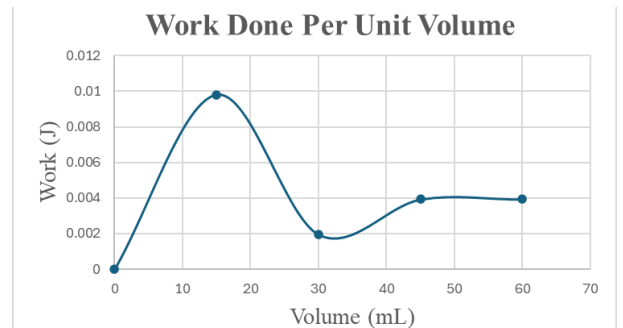


Figure 5: Work by Muscle with 200g Applied Load

III. ASSUMPTIONS AND APPROXIMATIONS

One assumption is that the actuators were symmetric and simply fabricated due to the provided materials, such as the heat press, it was difficult to uniformly seal the TPE.

IV. CONCLUSION

This lab introduced TPE as a soft robotic actuator material. TPE would be a preferred material over a silicone elastomer in any situation that requires an actuator that takes up the least amount of space when uninflated. TPE would also be preferred if there is a need for simple fabrication.